MIL-I-27709A(USAF)
13 September 1962
Superseding
MIL-I-27709 (USAF)
4 January 1962

MILITARY SPECIFICATION

INDICATOR, ATTITUDE ARU-14/A, REMOTE

1. SCOPE

1.1 This specification covers the requirements for one type of remote attitude indicator, designated ARU-14/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-L-6880	Lubrication of Aircraft, General Specification for
MIL-S-7502	Sealing Compound, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion, Accelerator Required
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for

FSC 6610

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MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8623	Adhesive, Epoxy Resin, Metal to Metal Structural Bonding
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-A-9067	Adhesive Bonding, Process and Inspection Requirements
MIL-I-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS24367	Lamp, Incandescent, Miniature, Integral Lighting
MS24515	Lamp, Sub Miniature
MS25237	Lamp, Incandescent, T-1-3/4 Bulb, Midget Flange Base
MS33545	Case - Instrument, 5 x 5-1/4 Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 COMPONENTS.- The indicator shall consist of servo mechanisms for actuating pitch and roll gimbals, an indicator display consisting basically of a fixed miniature airplane, bank angle dial, bank index, mechanism mask, moving two-colored spheroid background with a horizontal line dividing the two colors, a pitch trim knob and mechanism, external zeroing adjustments for pitch and roll, and a power-failure indicating device.

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143.

3.4 MATERIALS

3.4.1 FUNGUS-PROOF MATERIALS.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.4.2 NONMAGNETIC MATERIALS.- Nonmagnetic materials shall be used for all parts of the indicator except where magnetic materials offer definite advantages.

3.4.3 METALS.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.3.1 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion by means of a protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined by MS33586.

3.4.4 PROTECTIVE TREATMENT.- When materials are used in the construction of the remote indicator that are subject to deterioration when exposed to climatic and environment conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.4.5 CORROSIVE FUMES.- The materials as installed in the indicator and under the service conditions specified herein shall not liberate deleterious fumes.

3.5 DESIGN AND CONSTRUCTION.- The indicator shall be so designed and constructed that the complete mechanism can be removed from the case as a unit, and adjustments and repairs can be easily made by the personnel of operating units and overhaul bases. It shall be built to withstand the strains, jars, vibrations and other conditions incident to shipping, storage, installation, and service.

3.6 PERFORMANCE.- The indicator shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Operating temperatures ranging from -54° to +71° C and storage temperatures ranging from -62° to +71° C.

b. Relative humidity up to 100 percent

c. Pressures ranging from 57 inches Hg down to 0.81 inch Hg, approximately 80,000-foot altitude

d. Exposure to salt sea atmosphere

e. Vibration incident to service use

f. Fungus growth as encountered in tropical climates

g. Sand and dust particles as encountered in desert areas

h. Rainfall as encountered in any locale

i. Acceleration forces as anticipated

j. Immersion in water with pressure applied, or leak test conducted with an approved helium leak detector, without evidence of leakage exceeding that specified in 3.13.4

k. 500V dc for 10 seconds when applied as specified in 4.6.19.

3.6.1 RADIO INTERFERENCE.- The indicator shall be capable of meeting the radio interference requirements of MIL-I-26600.

3.6.2 LIFE.- The indicator shall be capable of meeting the performance requirements for 1,500 hours of operation during actual service usage.

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3.6.2.1 MEAN-TIME-BETWEEN-FAILURES (MTBF).- The indicator shall have a MTBF of 750 hours.

3.6.3 POWER CONSUMPTION.- Power consumption shall not exceed 20 va.

3.6.4 MAGNETIC EFFECT.- The indicator shall not cause deflection in excess of 5° when tested as specified in 4.6.17.

3.7 SERVOMECHANISMS.- The indicator shall be provided with two servomechanisms for actuating the pitch and roll gimbals.

3.8 INDICATOR DISPLAY.- The display shall be in accordance with figure 1 and shall consist basically of a fixed miniature airplane, a bank angle dial, a bank index, and a moving two-colored spheroid background with a distinct horizon line dividing the two colors. Unless otherwise specified, all colors shall be in accordance with FED-STD-595. The paint shall be such that the light will be sufficiently diffused to avoid highlight when the paint is illuminated with external or internal light sources, and markings will be clear and sharp when viewed under such conditions.

3.8.1 MINIATURE AIRPLANE.- The miniature airplane shall be in accordance with figure 2. The rear edge shall be within 1/16 inch of the spheroid surface. The wings shall follow the curvature of the spheroid with the top edge of the wings and center dot tilted back approximately 45° to facilitate night lighting. The miniature airplane shall be fixed directly to the coverglass at the bottom two points of the W. The attaching areas shall not exceed 1/8 inch diameter each and shall be color No. 37038. The W on the miniature airplane need not have a black border as shown on figure 2; however, the white portion of the W, wings, and dot shall be separated by an area of color No. 37038 of sufficient width to show contrast, subject to the approval of the procuring activity. The white areas of the miniature airplane shall be color No. 37875.

3.8.2 BANK ANGLE DIAL.- The bank angle dial shall be in accordance with figure 2. It shall be attached to the instrument case with the front edge (larger diameter) of the scale surface against the coverglass. The scale markings of the dial shall be color No. 37875.

3.8.3 BANK INDEX.- The bank index shall be in accordance with figure 2. It shall be affixed to and shall rotate with the roll gimbal but shall not move with pitch deviations. The front surface of the bank index shall be in a plane parallel to the coverglass, or beveled as approved by the procuring activity. If parallel, this plane shall be in the same plane as the front surface of the small diameter of the bank

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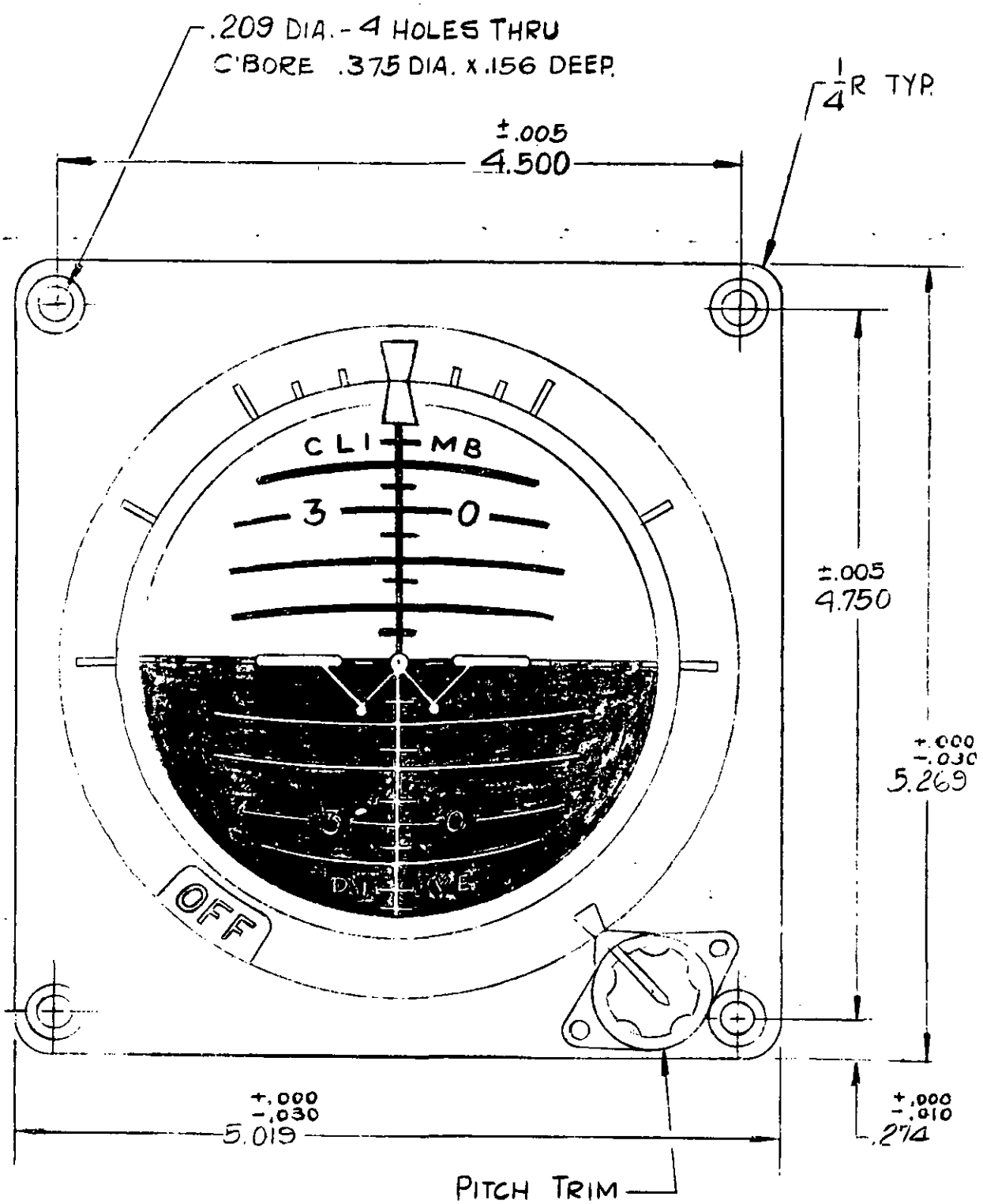
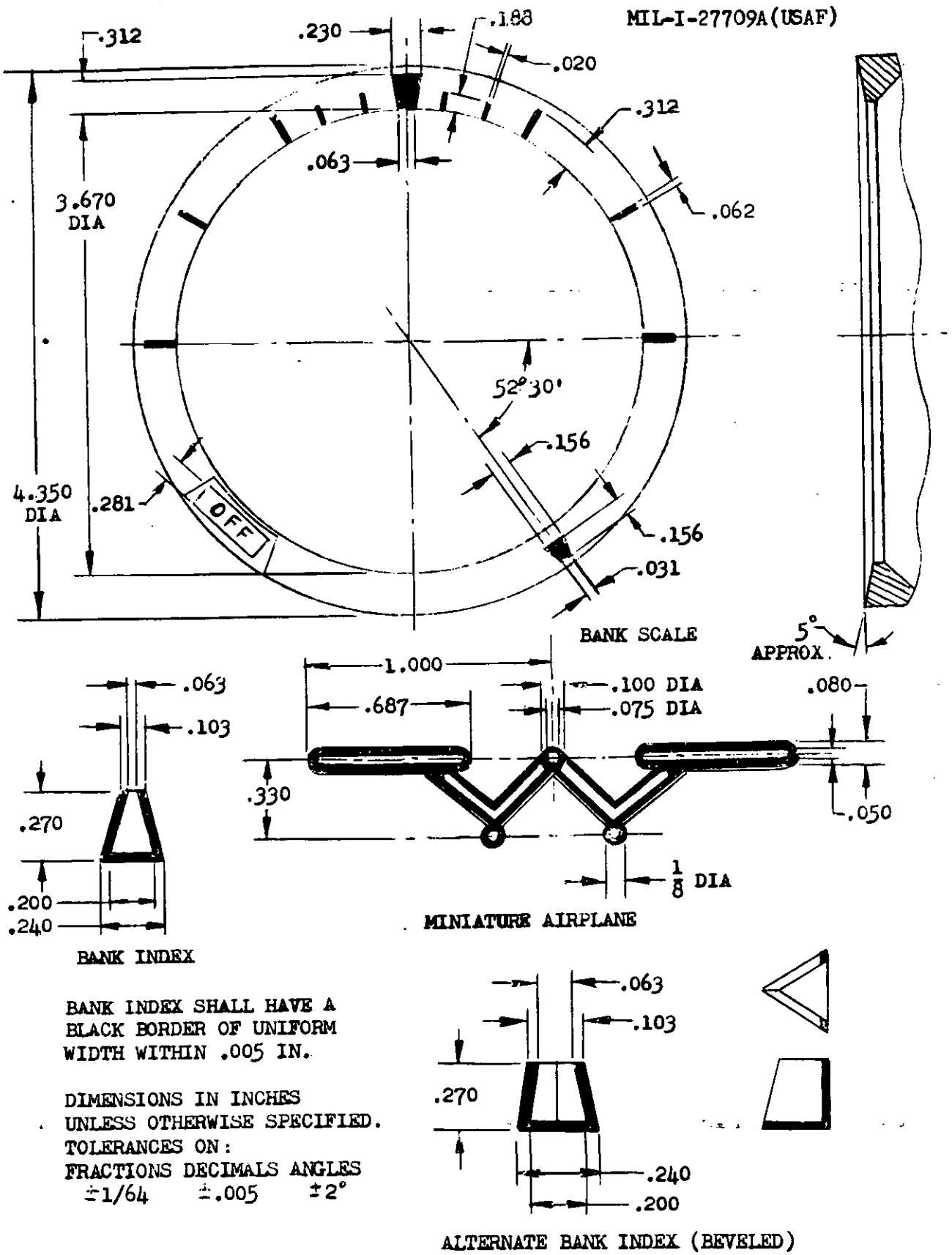


FIGURE 1 INDICATOR DISPLAY

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BANK INDEX SHALL HAVE A BLACK BORDER OF UNIFORM WIDTH WITHIN .005 IN.

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED.
TOLERANCES ON:

FRACTIONS	DECIMALS	ANGLES
±1/64	±.005	±2°

FIGURE 2. DIAL DISPLAY

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dial within ±3/64 inch. If beveled, the angles between the face of the index and the coverglass shall be approved by the procuring activity. The arc described by the foremost portion of the bank index shall not deviate more than 3/64 inch radially from the calibrated small diameter portion of the bank scale. At zero roll indication, the bank index shall be aligned with and radially within 3/64 inch of the zero roll marking on the bank scale. The index shall be color No. 37875.

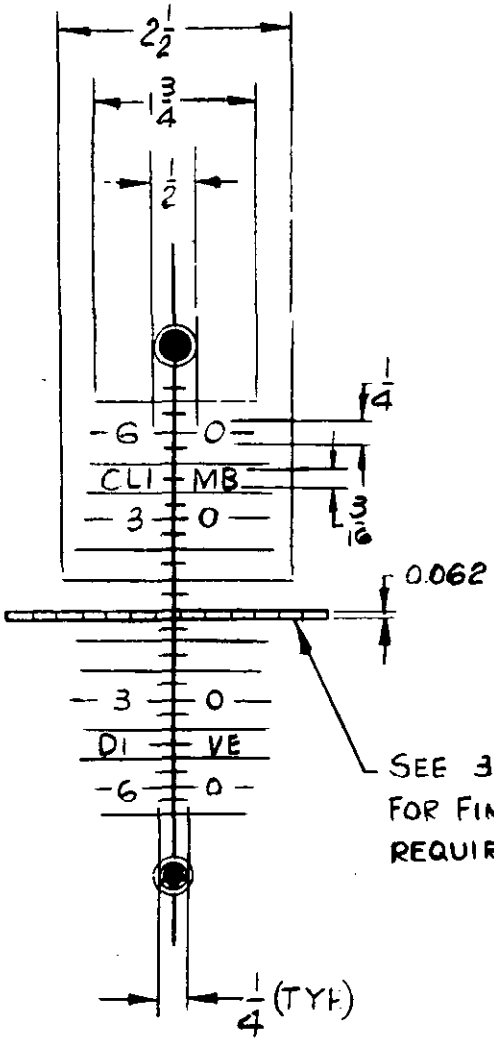
3.8.4 MECHANISM MASK.- A mask with the visible surface shaped as the surface of the frustrum of a cone shall be attached to the roll gimbal. The mask shall extend from the bank angle dial and shall be approximately 3-11/16 inches on the large diameter. This edge of the mask shall be hidden by the inner diameter of the bank angle dial. The smaller diameter of the mask shall be closest to the spheroid and shall intersect the spheroid in such a manner as to provide a lateral opening of at least 3-3/8 inches. This edge of the mask shall conform to the shape of the spheroid to within 1/16 ±1/32 inch.

3.8.5 SPHEROID.- The spheroid shall be mounted in the roll gimbal and shall rotate with the gimbal about the roll axis for roll indications and in relation to the roll gimbal, about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator and the pitch axis shall be horizontal when the indicator is level and indicating a zero roll attitude. The horizon line on the spheroid shall be fixed to the spheroid at the zero pitch position. The spheroid shall be driven in pitch so that during a dive maneuver, the horizon line will go above the miniature airplane and during a climb maneuver, the horizon line will go below the miniature airplane. The spheroid shall be in accordance with figure 3. The surface of the spheroid shall move 0.035 (nominal) inch for each degree of pitch change. The colors and markings shall be in accordance with figure 3 and the following requirements. The spheroid shall be internally illuminated.

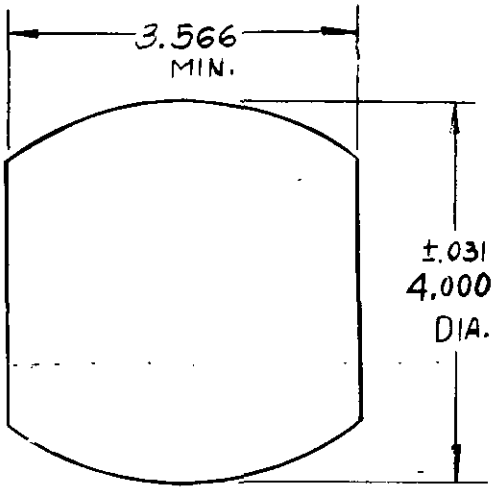
3.8.5.1 COLORS AND MARKINGS.- Above the horizon line, the spheroid shall be light gray, color No. 36373 and all markings shall be color No. 37038. The horizon line shall be alternately marked with colors No. 37038 and No.37875 as shown on figure 3. Below the horizon line, the spheroid shall be color No. 37038 and all markings shall be color No. 36440. The lengths of the major pitch lines shall be as follows:

0°	(Horizon line)	Width of sphere
10°		2-1/2 inches
20°		2-3/8 inches
30°		2-1/4 inches
40°		2-1/8 inches
50°		2 inches

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SEE 3.8.5.1 FOR FINISH REQUIREMENTS



SPHEROID
(FRONT VIEW)

ALL MARKINGS, NUMERALS AND LETTERS SHALL BE .020 WIDE UNLESS OTHERWISE SPECIFIED.

THE PITCH REFERENCE LINES SHALL BE PLACED ON THE SPHERE SUCH THAT THEY ARE PORTIONS OF GREAT CIRCLES WHICH INTERSECT AT THE PITCH AXIS.

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED
TOLERANCES ON
FRACTIONS ± 1/64
DECIMALS ±.005

FIGURE 3. SPHEROID AND MARKINGS.

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60° - 1-7/8 inches
70° - 1-3/4 inches
80° - Circle
85° - Solid dot

The center of the solid dots shall be the ±90° pitch position and from center to center shall represent 180° of pitch indication. The pitch indication shall be linear over the full 180° range which shall be spaced over 180° of the spheroid surface. The white areas shall be color No. 37875 and the black areas shall be color No. 37038.

3.9 PITCH TRIM KNOB.- A pitch trim knob shall be provided for adjusting the position of the spheroid. The knob shall be mounted on the lower right-hand side of the indicator and shall be marked as shown on figure 1. The knob shall be color No. 36231. The reference mark on the knob shall be color No. 37038, shall extend across the end of the knob, and shall continue along the side to a point on the knob that is adjacent to the triangular mark on the case at the zero pitch trim position. The triangular mark shall be color No. 37875. The knob shall have approximately 1/2 turn of freedom in the clockwise (cw) direction and 1/4 turn of freedom in the counterclockwise (ccw) direction from the pitch trim position. The freedom in each direction shall be limited by a position stop. When the knob is rotated to the stop in the cw direction, the spheroid shall rotate to deflect the horizon line upward to indicate between 10° and 20° dive. When the knob is rotated to the stop in the ccw direction, the spheroid shall rotate to deflect the horizon line downward to indicate between 5° and 10° climb. The servo system shall operate properly for any pitch trim adjustment. The knob shall have sufficient friction to remain fixed when set but it shall move smoothly without evidence of binding or erratic motion when being adjusted. The pitch trim circuit shall be so designed that if three ARU-11/A indicators are connected to a single test transmitter (pitch and roll), varying the pitch trim on any one of the indicators shall not affect the other indicators by more than 1/4° in the pitch and roll axis. A minimum of 2 and a maximum of 6 ounce-inches of torque shall be required to rotate the pitch trim knob through the full range of the settable indications in both the cw and ccw directions.

3.10 ZEROING ADJUSTMENTS.- Means shall be incorporated to externally adjust the position of the spheroid ±3° minimum in pitch and roll. The initial setting of the pitch and roll indication at the time of delivery of the indicator shall indicate zero pitch and roll within ±1/2° when the indicator is connected to a test transmitter set at electrical zero. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The roll adjustment shall be on the rear of the case

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and shall be adequately marked. It shall be equipped with stops and a locknut to secure the adjustment setting. The stops shall be so located that an erroneous indication of attitude cannot result from the positioning of the adjustment.

3.11 POWER FAILURE INDICATOR.- A power failure warning indicator shall be incorporated in the lower left of the indicator face on the bank angle dial as shown on figure 2. The word OFF shall be red on a white background. A lead shall be brought out through the electrical connector through pin J. The circuit shall be so designed that when pin J is connected to pin A and power is applied, the warning shall disappear. When the connection or the power is removed, or both, the warning shall appear. The red letters shall be illuminated under night lighting conditions. The lighting shall not be entirely dependent upon one light source.

3.12 PART NUMBERING OF INTERCHANGEABLE PARTS.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

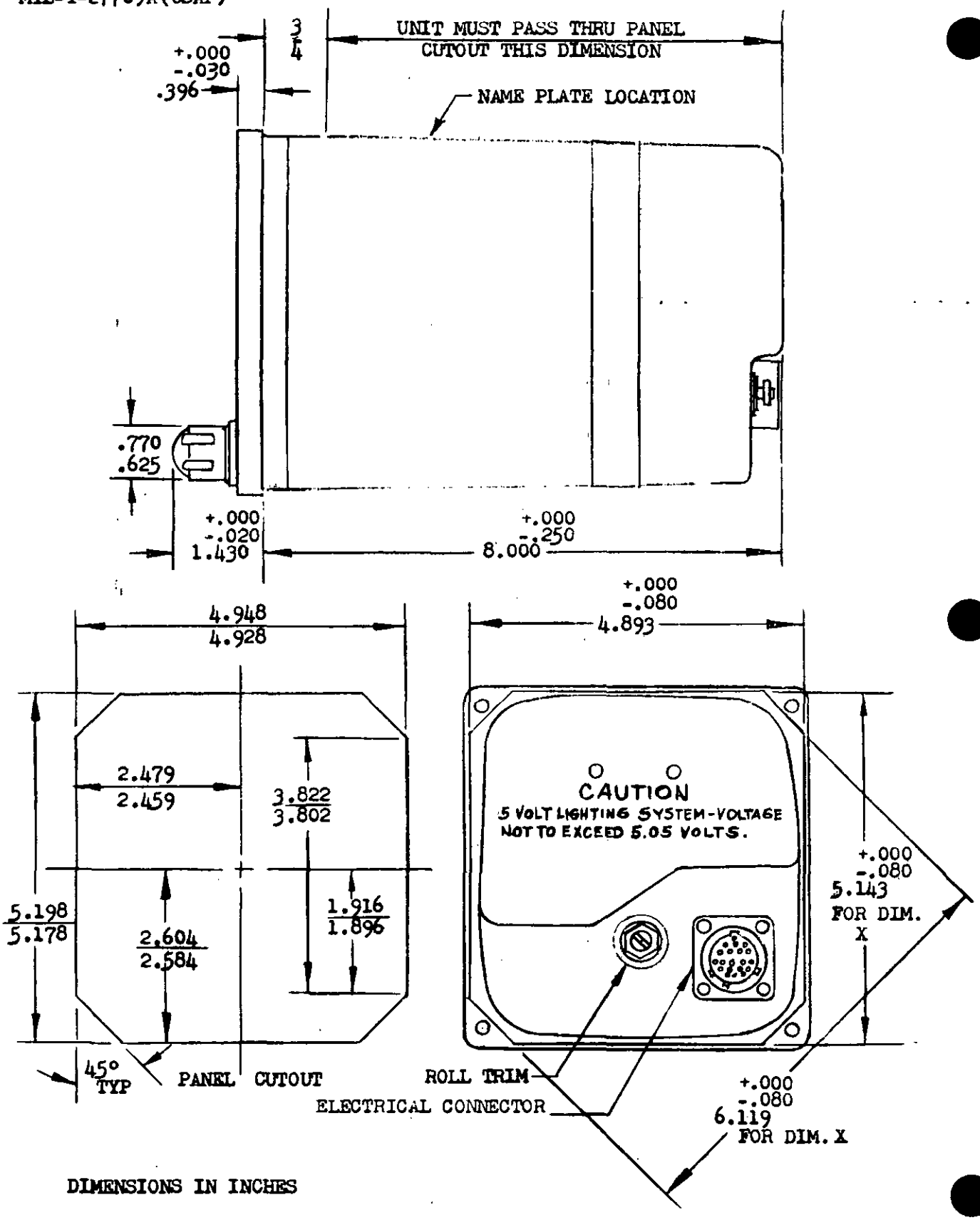
3.13 INDICATOR CASE.- The indicator case shall be hermetically sealed and in accordance with MS33545, except as shown on figures 1 and 4. The case shall be of sufficient strength to withstand an absolute pressure of 28 psi at a temperature of 25° C.

3.13.1 HERMETIC SEALING.- Hermetic sealing is the process by which the indicator is totally enclosed within a suitable metal structure or case and sealed airtight by fusion of metals by welding, brazing, or soldering; fusion of ceramic materials under heat or pressure; or fusion of ceramic materials into a metallic support. Elastomeric or resinous materials or combinations thereof, when used for the cover-glass seal, shall meet the leak rate requirements under all conditions specified herein. They shall be in accordance with MIL-S-7502, type B-2, or MIL-A-8623, type II or III. The surfaces to be bonded shall be cleaned in accordance with MIL-A-9067. Gaskets shall not be used to accomplish hermetic sealing.

3.13.2 COVERGLASS.- The coverglass shall be flat and free from discolorations, scratches, and striae that interfere with normal indicator reading. Its thickness shall not exceed 0.200 inch.

3.13.3 FILLING MEDIUM.- The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The filling medium shall be 100 percent helium, a mixture of 88 to 92 percent nitrogen and the remainder helium, or 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approximately 1 or 1/2 atmosphere, as required. Where practicable, the 100 percent helium filling medium shall be utilized.

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DIMENSIONS IN INCHES

FIGURE 4. INDICATOR CASE

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3.13.4 LEAK RATE.- The indicator shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

3.14 LIGHTING.- Integral lighting shall be provided. Supplemental lighting for the spheroid may be obtained by transillumination from within the spheroid. The integral lighting shall meet the requirements of MIL-L-25467. All production indicators shall be checked for proper lighting in accordance with 4.6.13.2. However, the gray portion of the spheroid shall at no time exceed 0.60 of the minimum brightness of the fixed white airplane.

3.14.1 EXTERNAL LIGHTING.- The indicator display shall be so designed that the display can be lighted by external means. This lighting capability shall be determined by use of Grimes fixture part No. B-4790, or equivalent.

3.14.2 PINS.- Power for the lighting design shall be in accordance with MIL-L-25467. Pins K and L shall be utilized for lighting purposes. The lighting lead, or leads, shall not be grounded to the indicator case.

3.14.3 RESISTORS.- Resistors shall not be used in the lighting system.

3.14.4 PRIME STANDARD.- Using parts that are identical to the parts to be used in production indicators, a lighted model of the indicator shall be constructed that meets all display requirements of MIL-L-25467, except for parts that affect the lighting design. The indicator model and recorded lighting data shall be submitted to the procuring activity for approval and, if approved, shall be certified as a prime standard.

3.14.4.1 PRIME STANDARD LIFE.- A minimum of 12 average areas covering all representative areas of the face of the prime standard shall be measured for brightness and color prior to submission and at least every 50 hours of operation thereafter. A change of ±20 percent in brightness or ±5 percent color in any two or more areas, using the initial data as a reference, shall void certification of the prime standard.

3.14.5 LAMPS.- Only the following lamps shall be used as a light source: MS24367-715AS15, MS24515-718AS15, and MS25237-328. Other lamps may be used provided they are approved by the procuring activity.

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3.15 ATTITUDE ACTUATION.- The spheroid shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the control transformer synchros on the indicator axes and the transmitter synchros in the pitch and roll displacement reference. The freedom of the spheroid rotation shall be unlimited about the roll and pitch axes.

3.15.1 SYNCHRO CONTROL TRANSFORMER.- The synchro control transformers on the pitch and roll axes shall be high-impedance synchros. When one to four indicators are connected to a single test transmitter per axis, no noticeable change in performance shall be noted.

3.15.2 STANDARD TEST TRANSMITTER.- The standard test transmitter shall be an Eclipse-Pioneer Type AY200-1, or equal, high-precision synchro with minimum and known errors. The rotor of the synchro shall be settable and rotatable through 360° and the settings shall be readable to within 6 minutes of arc. Each degree of scale marking shall represent 1° of synchro rotor rotation with a scale marked from 0° to 360° and so affixed to the synchro as to be readable as specified herein. The following method shall be used to calibrate the standard test transmitter:

a. The rotor leads shall be designated as H and C and one stator lead designated as Z and connected to C. When 26V is applied across H and C (C being ground) the scale shall be positioned on zero when the voltage across H and X is maximum and the voltage across the remaining two stator leads (X and Y) is minimum. The voltage across X and Z increases before it decreases for increasing numbers of the scale.

b. Synchro excitation - The test transmitter rotor shall be excited from the same power supply voltage that operates the indicator. It shall be so applied that the voltage from rotor lead C to rotor lead H is in phase with the voltage from pin A to pin B of the indicator connector with lead C and pin A common.

3.15.3 PITCH AND ROLL SENSING.- When the standard test transmitters, set at a scale reading of 180°, are connected to the indicator as shown in tables I and II, the indication shall drive to zero pitch and roll. Increasing scale setting of the test transmitters shall result in right bank and climb indications while decreasing the scale setting shall result in left bank and dive indications.

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TABLE I

Test Transmitter Terminal	Indicator Terminal of Power Input Connector	Rotor Excitation of Standard Test Transmitter
Hot (H) Cold (C)		26 V tap-off from single phase power supply to pins A and B of indicator
(X) (Y) (Z)	D E F	

TABLE II

Test Transmitter Terminal	Indicator Terminal of Power Input Connector	Rotor Excitation of Standard Test Transmitter
Hot (H) Cold (C)		26V tap-off from single phase power supply to pins A and B of indicator
(X) (Y) (Z)	R H P	

3.16 FOLLOWUP OPERATION.- The electrical and mechanical design of the indicator shall be such that in the process of followup, the indications in pitch or bank shall move smoothly without hunting, oscillating, or jumping more than 1/2° total amplitude in pitch or total amplitude in bank when connected to a standard test transmitter. The sustained electrical or mechanical oscillations in either axis shall be not more than 1/4° total amplitude at any indication. The servo mechanism shall be 70 to 100 percent critically damped and shall not lag the signal source by more then 10° at maximum rates of servo operation.

3.16.1 FOLLOWUP RATE.- The rate at which the motors drive the gimbals shall be not less than 300° per second in roll nor 180° per second in pitch.

3.16.2 FOLLOWUP ACCURACY.- The pitch and roll indications shall follow the input signals from their respective test transmitter synchros to within ±1/2° for positions between 0° and ±30° in roll and pitch and ±1° for all other positions.

3.17 AMPLIFIER CHANNELS.- The amplifiers shall be an integral part of the indicator. There shall be three transistor or magnetic amplifier stages. The output stage and at least one other stage shall be redundant. The indicator shall continue to operate with reduced performance should partial failure of the amplifier occur.

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3.18 ELECTRICAL POWER REQUIREMENTS.- The power supplied to the indicator shall be 115V, single-phase, 400 cps in accordance with MIL-STD-704. Power input connections shall be in accordance with table III.

TABLE III

Power Input Connections

Electrical Pins	Connections
A	Neutral (not grounded to case)
B	115V, 400 cps, single-phase power in phase with excitation to test transmitter synchros
C	No connection
(X)D (Y)E (Z)F	The three stator leads from the roll test transmitter synchro
(X)R (Y)H (Z)P	The three stator leads from the pitch transmitter
J	Power failure warning (connect to pin A to complete circuit)
K	
L	Internal lighting power input
M	Pitch null test terminal (pitch error voltage between this pin and pin A)
N	Roll null test terminal (roll error voltage between this pin and pin A)
G-S-T-U-V	No connection

3.18.1 ELECTRICAL POWER VARIATION.- The indicator shall operate properly and continuously at any a-c voltage between 103 and 127V and any frequency between 380 and 420 cps, or any combination thereof in addition to transient variations specified in MIL-STD-704.

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3.19 ELECTRONIC COMPONENTS.- The electrical components and applications thereof shall be in accordance with MIL-E-5400.

3.20 ELECTRICAL CONNECTOR.- The electrical connector shall be a Bendix Scintilla Type PT02H-14-19P, or equivalent, and shall be affixed to the case as shown on figure 4, except that it may be placed in the lower right-hand corner. This connector shall be hermetically sealed and shall mate with a Bendix Scintilla Type PT06P-14-19S, or equal. The pin connections shall be in accordance with table III. The electrical connector shall be equipped with a cap, or plug, and gasket in a manner that will protect the electrical surfaces from water vapor.

3.21 LUBRICATION.- The indicator shall be lubricated in accordance with MIL-L-6880.

3.22 MARKINGS.- Unless otherwise specified, the dimensions of all markings on the indicator shall be in accordance with figures 1 through 3 and all visible portions of the indicator shall conform to color No. 37038.

3.22.1 All markings shall be clear and sharp. Dimensions shall be as specified and on any given mark; the width shall not vary noticeably throughout the length of the mark unless otherwise specified. Markings which, for some given indication, become adjacent and are specified to have the same basic dimension at the adjacent point shall not be noticeably different at that point.

3.23 NUMERALS AND LETTERS.- Unless otherwise specified, all numerals and letters on the indicator, other than the nameplate, shall be in accordance with MS33558.

3.24 SOLDERING.- Soldering shall be accomplished in accordance with MIL-S-6872.

3.25 SCREW THREADS.- Unless otherwise specified, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.26 WEIGHT.- The weight of the indicator shall not exceed 5-1/2 pounds.

3.27 FINISHES AND PROTECTIVE COATINGS

3.27.1 ALUMINUM ALLOY PARTS.- Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.27.1.1 Dials, small holes, and case inserts need not be anodized.

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3.27.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.27.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 shall be used in lieu of anodizing.

3.27.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.27.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.27.1.6 Parts in a hermetically sealed container need not be anodized.

3.27.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.27.2 CADMIUM PLATING.- Cadmium plating shall be in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.28 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.29 WORKMANSHIP.- The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assembly screws and bolts.

3.29.1 RIVETING.- Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.29.2 CLEANING.- The indicator shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

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3.29.3 DIMENSIONS.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.29.4 SCREW ASSEMBLIES.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 SUPPLIER'S RESPONSIBILITY.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of the indicator shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values proper allowance shall be made for the difference from the specified condition.

4.3.2 VOLTAGE INPUT.- Unless otherwise specified, the indicator shall be tested with 115V 400-cps (nominal), single-phase power.

4.3.3 PROPERLY CONNECTED.- When the indicator is referred to as being properly connected, it shall be understood that it is connected to a test fixture that is providing a lighting circuit voltage variable from 5V (ac or dc) to approximately zero volts, synchro signals in accordance with 3.15.1 through 3.15.3, a-c power as required, and with pin J jumped to pin A through a closed switch.

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4.3.4 VISUAL OBSERVATION.- Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet.

4.4 QUALIFICATION TESTING

4.4.1 TEST SAMPLES.- The test samples shall consist of three indicators representative of the production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 QUALIFICATION TESTS.- The qualification tests shall consist of all tests specified under 4.6.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 INDIVIDUAL TESTS.- Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Starting

c. Pitch and bank zero

d. Pitch trim

e. Zeroing adjustments

f. Sensitivity

g. Followup rate

h. Followup accuracy

i. Gimbal freedom

j. Leak rate

k. Power warning indicator

l. Fogging

m. Lighting

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4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A.- One indicator shall be selected at random from every 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Power consumption

c. Power variation

d. Followup operation (except damping)

e. Magnetic effect

f. Acceleration (with power applied)

g. Dielectric strength

h. Vibration error

i. Horizon line and bank index null position shift

j. Low temperature operation

k. High temperature operation

4.5.2.2 SAMPLING PLAN B.- Unless otherwise specified, 3 indicators selected at random from the first 15 items on the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Radio noise interference

c. High temperature exposure

d. High altitude - low temperature

e. Temperature shock

f. Humidity

g. Fungus

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h. Rain

i. Salt spray

j. Sand and dust

k. Acceleration (without power)

l. Life test

m. Vibration failure

n. Damping

o. Pressurization test.

NOTE: At the option of the procuring activity, the tests may be divided into three groups as follows, to be conducted on three sets of three instruments each, if a need exists, to expedite tests. Each indicator shall meet the requirements of all the individual tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II may be conducted on empty cases that have all external parts installed or attached as on a completed indicator.

Group I	Group II	Group III
Radio noise interference	Humidity	Vibration failure
High temperature exposure	Fungus	Acceleration
Low temperature	Rain	Life
Temperature shock	Salt spray	
	Sand and dust	

4.5.2.3 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary test shall be repeated.

4.5.2.3.1 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

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4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 STARTING.- The indicator shall be properly connected and power applied. The warning flag shall disappear and the indicator shall be ready for immediate operation.

4.6.3 PITCH AND BANK ZERO.- The standard test transmitters in the pitch and roll axes shall be set in the electrical zero position. Power shall be applied and the indicator shall followup to the zero indications in pitch and bank within 1/2° from any random indication in either axis. This test shall be repeated and the indication shall return to within 1/4° of the original indication.

4.6.4 PITCH TRIM.- The pitch trim knob shall be adjusted to its limits of 1/4 turn ccw and 1/2 turn cw and the deflection of the horizon line observed. The respective deflections shall be from 5° to 10° down and from 10° to 20° up from the level flight indications. With the pitch trim knob in the zero pitch trim position, the indicator shall indicate zero pitch within 1/2 of the horizon line width. The torque required to rotate the pitch trim knob shall be measured throughout the range of settable indications in both the cw and ccw directions and shall be within 2 to 6 ounce-inches. The ARU-14/A indicators shall be connected to the same transmitter and the pitch trim knob moved through its range of motion on one of the indicators. There shall be not more than 1/2° change in the indication of the other indicator.

4.6.5 ZEROING ADJUSTMENTS.- The indicator shall be properly connected to the standard test transmitters with each set to electrical zero. The roll zeroing adjustment shall be moved to its extremes and the deflection of the bank index noted. This deflection shall be not less than ±3° from the zero position. The pitch test transmitter shall be rotated 3° in climb then 3° in dive. In each case, the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zeroed. The pitch trim knob shall be tightened again to the shaft at the zero position. In each case, the requirements of 4.6.4 shall be met, except the last two sentences. Where sampling plan B tests specify repetition of the individual test, this test shall not be repeated except as part of the life and MTBF test.

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4.6.6 SENSITIVITY.- The test transmitter for the roll axis shall be moved 1/4° in right bank and returned to the original position, 1/4° in the left roll and returned to the original position. The test transmitter for the pitch axis shall be moved 1/4° in dive and returned to the original position then 1/4° in climb and returned to the original position. There shall be a smooth, perceptible movement of the indicator bank index and horizon line for each movement of the appropriate test transmitter. The test shall be repeated in dive and climb with the pitch trim in each of its extreme positions, except the test transmitter shall be moved 1/2° in each direction.

4.6.7 FOLLOWUP RATE.- The indicator shall be properly connected to a remote signal source or standard test transmitter. The followup rates in pitch and roll shall be determined. The computed rate in the roll axis shall be not less than 300° per second and the computed rate in the pitch axis shall be not less than 180° per second.

4.6.8 FOLLOWUP ACCURACY.- The rotor of the standard test transmitter in the roll axis shall be displaced to at least five random positions other than the zero position. The indicator shall position itself with the rotor of the standard test transmitter within ±1/2° for each position between 0° and 30° left and right bank and within ±1.0° for any position over the remaining 300°. The rotor of the standard test transmitter in the pitch axis shall be displaced to at least five random positions other than the zero position. The indicator shall position itself with the rotor of the standard test transmitter within ±1/2° for each position between 0° and 30° dive or climb and within ±1.0° for any position greater than ±30°.

4.6.9 GIMBAL FREEDOM.- The indicator shall be properly connected and the remote signal source or standard test transmitter moved 360° in roll and pitch. The indicator shall follow smoothly without sticking and overshooting or hunting greater in magnitude than that outlined in 4.6.16.2.

4.6.10 LEAK RATE.- The instrument case shall be tested for leaks by means of a mass spectrometer type of helium leak detector or other suitable means approved by the procuring activity. The maximum leak rate shall not exceed that specified in 3.13.4.

4.6.11 POWER WARNING INDICATOR.- The indicator shall be properly connected with no power applied. Power shall be applied and the flag shall disappear. The power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from pin J to A disconnected; the flag shall appear.

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4.6.12 FOGGING.- The indicator shall be energized and maintained in a 71° ±2° C controlled ambient for a minimum period of 1 hour. After this period and while at the specified temperature, an ice cube shall be rubbed on the indicator glass face for a period of 30 ±5 seconds. The glass shall be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.13 LIGHTING

4.6.13.1 PRIME STANDARD AND QUALIFICATION.- Lighting of indicators to be used as prime standards and indicators submitted for qualification shall be tested as specified in MIL-L-25467.

4.6.13.2 PRODUCTION TESTING.- Lighting tests on production indicators shall be accomplished as follows. Particular attention shall be given to individual segments of the indicator when making the specified comparisons.

a. Apply 5V to the prime standard and to the production indicator being checked. The two indicators shall be visually compared and shall be identical in color.

b. Apply 3V to the prime standard and 3.2V to the indicator being checked. The instrument being checked shall visually appear no brighter than the prime standard.

c. Apply 3V to the prime standard and 2.8V to the indicator being checked. The instrument being checked shall visually appear no dimmer than the prime standard.

4.6.13.2.1 Any indicator that does not meet the visual comparison tests specified in 4.6.13.2 shall not be approved unless the indicator is submitted to all display tests listed in MIL-L-25467 and sufficient data is taken to substantiate conformance to the display requirements therein.

4.6.14 POWER CONSUMPTION.- The power consumption of the indicator shall be measured with suitable meters and shall not exceed 20va. The power factor shall be maintained between 0.85 and 1.00 lag.

4.6.15 POWER VARIATIONS.- With a-c power applied, the indicator shall be properly connected and subjected to the transient power conditions of MIL-STD-704. There shall be no damage to the indicator. The indicator shall then be subjected to the tests specified in 4.6.6, except the tolerance shall be 1° in roll and 3/4° in pitch, 4.6.3, except the bank zero tolerance shall be 3/4°, 4.6.9, and 4.6.11 at each of the following voltage and frequency settings:

a. 103V - 380 cps

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b. 127V - 380 cps

c. 103V - 420 cps

4.6.16 FOLLOWUP OPERATION.- Suitable test equipment shall be connected to the appropriate test pins of the indicator and any hunting, jumping, lag, or overshooting of the indication when the indicator is made to follow the standard transmitter at various constant rates in both the pitch and roll axes observed and recorded. This shall be possible with the indicator sealed. While performing the individual tests, hunting, jumping, and oscillation shall not exceed ±1/8°. Pen recorders shall not be used to determine followup rates and oscillation.

4.6.16.1 INDICATION LAG.- In the pitch axis, the indication shall not lag behind the pitch remote input transmitter by more than 5° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 180° per second. In the roll axis, the indication shall not lag behind the roll remote input transmitter by more than 4° for a shaft speed of 90° per second nor more than 10° for a shaft speed of 300° per second.

4.6.16.2 HUNTING AND JUMPING.- There shall be no noticeable hunting and jumping of the indicator display in pitch or roll with no rate inputs from the test transmitters. The test transmitter shafts shall be rotated separately through all rates from zero through the maximum speed of the respective axis of display. The survey of rates shall be sufficiently slow to record the hunting and jumping that occurs at any speed of each axis of the indicator display and shall take not less than 1 minute with the rate varying approximately linearly. The peak-to-peak amplitude of the hunting and jumping shall not exceed 1/8° for shaft speeds between 0° and 20° per second in pitch and roll, 1/2° for shaft speeds between 20° and 90° per second in pitch and roll, nor 1° for shaft speeds between 90° and 300° per second in roll.

4.6.16.3 DAMPING.- The damping of the indicator servos shall be determined and shall be 70 to 100 percent of critical.

4.6.17 MAGNETIC EFFECT.- This test shall be made first with no power applied to the indicator and then with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and 5 inches from the center of a free magnet, approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted, and the maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

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4.6.18 ACCELERATION (WITH POWER APPLIED).- The indicator shall be properly connected and power applied to the attitude circuit. The indicator shall be accelerated along its normal vertical axis with a force of 5g. The attitude display shall not vary by more than 1° in pitch or roll from the initial indication. This test shall be conducted with the attitude set at zero pitch and roll and with each axis displaced 90°.

4.6.19 DIELECTRIC STRENGTH.- A potential of 500V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.20 HORIZON LINE AND BANK INDEX NULL POSITION SHIFT.- The indicator shall be properly connected to standard test transmitters in the pitch and roll axes and the transmitter set to drive the indications in both axes to the zero position. The input voltage and frequency shall be varied within the following limits and combinations:

Frequency	380 to 420 cps
A-C voltage	103 to 127V

The horizon line shall not shift more than ±1/16 inch from normal null position. The bank index shall not shift more than 1/2 of the narrow end of the point width from normal null position.

4.6.21 ENVIRONMENTAL TESTS.- The indicator shall be subjected to the following environmental tests, conducted in accordance with the applicable procedures of MIL-E-5272 and as specified:

4.6.21.1 VIBRATION ERROR.- The indicator shall be mounted as specified in procedure IV, properly connected, and subjected to vibration with a total amplitude between 0.003 and 0.005 inch. The indicator shall meet the following requirements:

4.6.21.1.1 ROLL AND PITCH OSCILLATION.- The maximum total amplitude of roll and pitch indicated oscillation throughout the frequency range shall be noted and shall not exceed 1/2°.

4.6.21.1.2 ROLL AND PITCH VARIATION.- During vibration, the maximum variation of the roll and pitch indication from the original indication before vibration, throughout the frequency range, shall be noted and shall not exceed 1/2°.

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4.6.21.2 LOW TEMPERATURE OPERATION.- The indicator shall be properly connected, with no power applied, and placed in a chamber at a temperature of -54° ±2° C for a period of 4 hours. At the end of this period and while the indicator is at the reduced temperature, power shall be applied for a period of 10 minutes. While still at the reduced temperature, the indicator shall be subjected to and shall meet the tests specified in 4.6.13, except the tolerance shall be ±1°, and 4.6.6, except the tolerance in pitch and roll with the pitch trim knob in the zero position shall be ±1/2°, and the tolerance in dive and climb with the pitch trim knob in each of its extreme positions shall be ±1°.

4.6.21.3 HIGH TEMPERATURE OPERATION.- The indicator shall be properly connected, placed in a chamber at a temperature of +71° ±2° C, and operated with power applied for a period of 4 hours. At the end of this period and while at the high temperature, the indicator shall meet the tests specified in 4.6.3, except the tolerance shall be ±1°, and 4.6.6, except the zero pitch trim tolerance shall be ±1/2° and ±1° with the pitch trim knob at the extreme positions.

4.6.21.4 HIGH TEMPERATURE EXPOSURE.- The indicator shall be properly connected with no power applied and subjected to a high temperature exposure test in accordance with procedure II for a period of 24 hours. At the end of this period and while still at the high temperature, the indicator shall meet the test specified in 4.6.9.

4.6.21.5 HIGH ALTITUDE-LOW TEMPERATURE.- The indicator shall be subjected to a high altitude-low temperature test in accordance with procedure II. The chamber shall be reduced to -62° C with the indicator properly connected but no power applied. After a period of 48 hours, the temperature shall be raised to -54° C for an additional 24 hours. At the end of this period, the pressure in the chamber shall be reduced to the equivalent of 80,000 ±500 feet and power applied. The time required for the indicator to start operating shall not exceed 5 minutes. There shall be no binding or sticking at any indication. After the indicator has returned to room temperature and the pressure has increased to atmospheric, it shall be subjected to and shall meet the individual tests.

4.6.21.6 TEMPERATURE SHOCK.- The temperature shock test shall be conducted in accordance with procedure I. At the conclusion of this test, the indicator shall be subjected to and shall pass the individual tests.

4.6.21.7 HUMIDITY.- The humidity test shall be conducted in accordance with procedure I. There shall be no sticking or binding in the pitch trim knob nor evidence of corrosion or rusting of internal or external parts as a result of this test.

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4.6.21.8 FUNGUS.- The fungus test shall be conducted in accordance with procedure I. There shall be no sticking or binding in the pitch trim knob nor deterioration that would adversely affect subsequent operation. Fungus growth shall not be supported as a result of this test.

4.6.21.9 RAIN.- The rain test shall be conducted in accordance with procedure II, except that the indicator shall be mounted for 1/2 hour with the dial in a vertical plane and 1/2 hour with the dial in a horizontal plane (dial face upward). Upon removal from the chamber, there shall be no evidence of sticking or binding in the pitch trim knob as a result of this test.

4.6.21.10 SALT SPRAY.- The salt spray test shall be conducted in accordance with procedure I. At the end of the 50-hour period, there shall be no sticking or binding of the pitch trim knob or damage to the indicator. The tolerances specified under the individual tests shall apply.

4.6.21.11 SAND AND DUST.- The sand and dust test shall be conducted in accordance with procedure I. At the end of this test, the indicator shall be removed and allowed to cool at room temperature. There shall be no sticking or binding of the pitch trim knob.

4.6.21.12 VIBRATION FAILURE.- The vibration failure test shall be conducted in accordance with procedure V. At the conclusion of this test, the indicator shall meet the tests specified for pitch and bank zero and sensitivity. No looseness in the mechanism nor damage to any part of the indicator shall result from this test.

4.6.22 PRESSURIZATION TEST.- The indicator shall be placed in a chamber at an absolute pressure of 57 inches Hg and a temperature of 25° C and maintained under these conditions for a period of 4 hours. At the end of this period, the indicator shall be closely examined and any evidence of deformation, damage, or unsatisfactory operation shall be cause for rejection. The indicator shall then be subjected to and shall pass the test specified in 4.6.10.

4.6.23 ACCELERATION.- The indicator, not operating, shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for a period of 1 minute. At the end of this acceleration, the indicator shall meet the individual tests. No damage to the indicator shall result from this test.

4.6.24 RADIO NOISE INTERFERENCE.- The radio noise interference test shall be conducted in accordance with MIL-I-26600.

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4.6.25 LIFE TEST.- The indicator shall be properly connected and subjected to a cycle consisting of a 4-hour period of operation followed by a period of 5 minutes during which no power shall be applied to the indicator. During the operating periods, both axes of the attitude display shall be oscillated through a minimum of ±30° at a rate of 5 to 7 times per minute, the lighting circuit shall be energized with 5V ac or dc, and the power warning flag system shall show that the indicator is in operation. During the resting or cooling-off periods, all power shall be removed from the instrument under test, including the power to the lighting and flag circuits, and the standard test transmitter shall remain excited and its rotor shall continue to be displaced through a minimum of ±30° at a rate of 5 to 7 times per minute. At least once during each 4-hour period, the pitch trim knobs shall be rotated to their extremes 10 times in each direction. At the end of 1,500 hours, the pitch trim torque shall be between 2 and 6 ounce-inches. The above cycle shall be repeated a sufficient number of times to extend the operating time of the indicator to a total of 1,500 hours. There shall be no failure of the indicator during this test except as limited by 4.6.25.1. Hours accumulated during the environmental tests, except for group II tests, may be considered as part of the 1,500-hour life test. After the life test, the indicator shall meet the sampling plan A tests.

4.6.25.1 MTBF.- Upon completion of all tests specified in section 4, a MTBF shall be calculated, exclusive of group II tests. The MTBF shall be not less than 750 hours. In the event a MTBF of 750 hours cannot be verified, three production units shall be picked at random and subjected to the life test. The calculated MTBF for the retest samples shall be not less than 900 hours. In the event 900 hours MTBF cannot be established, 4.5.2.3 shall apply. MTBF shall be considered as the total operating time accumulated on all test units divided by the number of failures. All out-of-tolerance conditions shall be considered as failures. For the purpose of this test, an out-of-tolerance condition shall be defined as follows (applicable to 4.6.25 only):

a. Where a tolerance-type requirement is specified, a failure shall be considered as any electrical or mechanical condition which results in an increase of 100 percent or more of the specified tolerance.

b. Where a maximum- or minimum-type requirement is specified, a failure shall be considered as any electrical or mechanical condition which results in a value which is greater then the specified maximum or less than the specified minimum by more than 5 percent.

c. Where a condition is specified as a requirement, a failure shall be considered as any physical, electrical, or mechanical condition which fails to comply with the requirements specified herein.

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5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVELS A, B, AND C.- Indicators shall be preserved in accordance with method III of MIL-P-116 and packaged in accordance with the applicable levels of MIL-P-7936.

5.2 PACKING

5.2.1 LEVELS A AND B.- Indicators preserved and packaged as specified in 5.1.1 shall be packed in accordance with levels A and B of MIL-P-7936.

5.2.2 AIR SHIPMENT.- Indicators shall be packed in accordance with level B of MIL-P-7936.

5.3 MARKING FOR SHIPMENT.- Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 INTENDED USE.- The ARU-11/A remote attitude indicator covered by this specification is intended for use in high performance aircraft. The indicator is controlled by a vertical reference control which provides transmitting synchro pitch and roll output signals.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Whether sampling plan B tests are to be conducted

c. Selection of applicable levels of packaging and packing.

6.3 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, ATTN: ASNPFC, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

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